

COMPUTER SECURITY (320)

STATE – 2019

TOTAL POINTS _____ ***(500 points)***

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than sixty (60) minutes testing time

Property of Business Professionals of America.
May be reproduced only for use in the Business Professionals of America
Workplace Skills Assessment Program competition.

MULTIPLE CHOICE

Identify the letter of the choice that best completes the statement or answers the question. Mark "A" if the statement is true. Mark "B" if the statement is false.

- 1) What does the acronym WEP stand for?
 - a) Wired Equivalent Privacy
 - b) Wireless Encryption Protocol
 - c) Wifi Enhancement Process
 - d) Wireless Entrance Procedures
- 2) WannaCry is an example of what kind of attack?
 - a) DDOS
 - b) Ransomware
 - c) Polymorphic
 - d) Fork bomb
- 3) In order to protect against potential attacks, what could you do to protect a web server?
 - a) Use a firewall to block port 80
 - b) Replace Microsoft IIS with Apache
 - c) Change the port address to 1019
 - d) Web server cannot be secured
- 4) Which mechanism is used by PKI to allow immediate verification of a certificate's validity?
 - a) OCSP
 - b) CRR32
 - c) SHA256
 - d) CRL
- 5) Fuzzing is the name for a method that does which of the following?
 - a) Inserting unexpected values as input into an application to try and break it
 - b) Creating multiple attack vectors to see which one works against a system
 - c) Confusing a web application by spoofing your IP address multiple times
 - d) Performing a fuzzy search on a list of passwords to find the correct on
- 6) The primary organization or body for maintaining certificates is called the _____.
 - a) CSA
 - b) CA
 - c) IRL
 - d) LRA

- 7) SLA (service level agreements) contain negotiations for which of the following services?
- a) Logging
 - b) Auditing
 - c) Regulatory compliance
 - d) All of the above
- 8) The word "BPA" when run through a rot14 cypher would spell...
- a) NCO
 - b) OCN
 - c) CQB
 - d) APB
- 9) Cybercrime falls under the jurisdiction of which U.S. federal agency?
- a) Secret Service
 - b) Police
 - c) FBI
 - d) NSA
- 10) What does the "A" in the information security CIA triad stand for?
- a) Atomic
 - b) Authenticated
 - c) Accessibility
 - d) Availability
- 11) Which of the following is the mask for a Class A IP address?
- a) 255.255.255.0
 - b) 255.1.1.0
 - c) 255.0.0.0
 - d) 255.255.0.0
- 12) What is the name of the technology that is currently being developed that has the potential to revolutionize the financial industry?
- a) Bitchain
 - b) Blockchain
 - c) GraphConnect
 - d) Listchain
- 13) What does PCISS stand for?
- a) Payment Card Industry Data Security Standard
 - b) Payment Creation and Initiation Data Security Standards
 - c) Personal Computer Information Data Security Set
 - d) Platform for Creating Independent Data Sets and Software

- 14) What is “data about data” called?
- a) Hyperdata
 - b) Systemic Data
 - c) Fuzzy Data
 - d) Metadata
- 15) A network mapper is *unable* to do which of the following?
- a) Determine the physical location of a device on a network
 - b) Find unknown devices on a network
 - c) Find all the IP addresses currently in use on the network
 - d) Provide some information on the services that certain hosts on the network provide
- 16) _____ are the *most* vulnerable asset that are prone to an attack within an organization.
- a) People
 - b) Network switches
 - c) Routers
 - d) Servers
- 17) What is the name of the technology that allows a connection to be made between two computers on disparate networks?
- a) Tunneling
 - b) Subnetting
 - c) Blockchain
 - d) Extranet
- 18) An alert signals the network administrator that a server on the network has a program running on it that has bypassed authorization. What type of attack has occurred?
- a) Distributed denial of service
 - b) Social engineering attack
 - c) Denial of service
 - d) Backdoor
- 19) RBAC, regarding access control, stands for _____.
- a) Role Based Access Control
 - b) Residual Binary Access Control
 - c) Role Balanced Access Control
 - d) Rule Based Access Control
- 20) Which of the following is an example of biometric security?
- a) Retinal scan
 - b) Fingerprint scan
 - c) Facial recognition
 - d) All the above

- 21) Company A occupies a three-story building in downtown Austin, TX. The data that the company deals with is highly confidential and is quite attractive to hackers and potential attackers. The company is wanting to build a brand new datacenter to complement their existing cloud infrastructure. What floor is the best floor to put the datacenter on?
- a) First floor
 - b) Second floor
 - c) Third floor
 - d) Construct a basement to house the datacenter
- 22) Which of the following is a protocol used to facilitate the delivery of email?
- a) FTP
 - b) SMTP
 - c) AES
 - d) SSH
- 23) Which of the following protocols is used in network to prevent switching loops?
- a) AES
 - b) RSA
 - c) STP
 - d) UDP
- 24) Which act enforces the need for strict information security policies in the financial industry?
- a) FISA
 - b) GLBA
 - c) SOX
 - d) FERBA
- 25) When an email is crafted to make it look benign, but is in fact used to lure a victim into a malicious attack or identity theft, this is known as?
- a) Phreaking
 - b) SPAM
 - c) Phishing
 - d) Trojan horse
- 26) What does a load balancer do?
- a) Distribute requests to multiple different servers
 - b) Manage a queue of service tickets
 - c) Balance network load across multiple ports on a physical switch
 - d) A physical assembly that balances the load of server racks in a datacenter
- 27) Information security is the responsibility of _____.
- a) Network administrators
 - b) The information security office
 - c) Everybody
 - d) Upper management

- 28) What port is used by HTTP?
- a) 80
 - b) 25
 - c) 43
 - d) 21
- 29) What kind of attack is performed when an attacker tries to gain access to a system by repeatedly guessing the password?
- a) Rainbow attack
 - b) Brute force attack
 - c) Triggering attack
 - d) Fuzzer attack
- 30) Which of the following is a form of ransomware?
- a) Crypto-malware
 - b) Systemic Ransom
 - c) Port Holder
 - d) Port blocker
- 31) How many layers are in the Open Systems Interconnection model?
- a) 4
 - b) 5
 - c) 6
 - d) 7
- 32) What is the super user account on a Linux operating system called?
- a) Super User
 - b) Master
 - c) Root
 - d) Administrator
- 33) Which of the following is a type of MITM attack?
- a) CryptoWare
 - b) SSLStrip
 - c) Low Orbit Ion Cannon
 - d) Conficker
- 34) Which of the following is a tool used for stress testing networks, but is also used to DoS or DDoS a network host?
- a) Low Orbit Ion Cannon
 - b) Port Cannon
 - c) Port Blaster
 - d) Network Denier

- 35) When a program is changed manually to improve something or to enhance security, it is called applying a _____.
a) Fix
b) Hotfix
c) Patch
d) Delta
- 36) Outer-level access control in a physical security system is addressed by what component?
a) Mantraps
b) Doors
c) Scanners
d) Perimeter security
- 37) Which of the following is the name of the hypervisor provided by Microsoft?
a) HyperV
b) VirtualBox
c) VMWare
d) Centrix
- 38) What is the name of Apple's zero-configuration network service?
a) Hello
b) Bonjour
c) MDNS
d) DDNS
- 39) What attack takes advantage of the probability that two keys match?
a) Rarity attack
b) Rainbow attack
c) Birthday attack
d) Uniqueness attack
- 40) Of the following backup types, which one will *most* likely provide the quickest backup time?
a) Full backup
b) Delta backup
c) Archival backup
d) Incremental backup
- 41) What tool is used in Windows to encrypt an entire volume?
a) BitLocker
b) BitDefender
c) Norton Antivirus
d) Windows Firewall

- 42) What is an attack that targets DNS data?
- a) DNS spotting
 - b) DNS re-routing
 - c) DNS cache poisoning
 - d) DNS resolution redirection
- 43) What does DNS stand for?
- a) Data Network Solution
 - b) Distributed Name Solving
 - c) Domain Network Software
 - d) Domain Name System
- 44) Which of the following is a language used to interact with a relational database management system?
- a) Gremlin
 - b) Cypher
 - c) BSON
 - d) SQL
- 45) What is the super user account on a Windows operating system called?
- a) Super User
 - b) Master
 - c) Root
 - d) Administrator
- 46) Information security falls solely on the information security officer.
- a) True
 - b) False
- 47) Routers are the most vulnerable component when it comes to information security.
- a) True
 - b) False
- 48) Worms aim to spread to other devices by means of network connections.
- a) True
 - b) False
- 49) Phreaking is a process by which harmful emails are made to seem legitimate.
- a) True
 - b) False
- 50) DNS associates human-readable names with IP addresses.
- a) True
 - b) False

